TITLE PAGE

Problem Statement ID – SIH26022

Problem Statement Title – Design and develop a smart, solar-powered drying and compact packaging system to support home-based agarbatti manufacturing by rural women artisans

Theme – Agriculture, FoodTech & Rural Development

PS Category – Hardware

Team ID – 128479

Team Name (Registered on portal) – Quantum Loop

Quantum Loop

IDEA TITLE

SMART INDIA HACKATHON 2026

SOLKIRAN — Smart Solar Drying and Packaging System for Rural Artisans

Proposed Solution

- A solar-hybrid powered drying and packaging unit for home-based agarbatti manufacturing, replacing inconsistent open-air sun drying with a controlled, automated, hygienic process
- Artisan loads rolled agarbatti sticks into a compact solar drying chamber; the system handles drying, sterilization, moisture-based cutoff, and semi-automated packaging with minimal manual monitoring

Innovation & Uniqueness

- Hybrid solar + battery power – stores excess daytime energy so drying continues on cloudy days or after sunset
- UV-C sterilization stage – kills bacteria and fungal spores before packaging for hygienic, export-quality output
- Moisture-triggered auto-cutoff – humidity sensors stop drying at the exact optimal moisture level, no fixed timer
- Companion mobile app – multilingual, voice-friendly; remote batch start, live status, batch-ready notifications
- QR-coded batch traceability – each pack links to batch date, drying conditions and artisan profile

Batch Drying & Packaging Flow

1. Artisan loads rolled agarbatti sticks into drying chamber
2. Hybrid solar/battery power runs heater and airflow system
3. UV-C sterilization cycle runs inside the chamber
4. Humidity sensor monitors moisture and auto-cuts off at optimal level
5. Companion app notifies artisan that the batch is ready
6. Semi-automated packaging with QR-coded batch label generated



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TECHNICAL APPROACH

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Technologies to Be Used

- Solar panel with battery backup and charge controller (hybrid power system)
- Microcontroller (ESP32/Arduino) for sensing and control logic
- Temperature and humidity sensors for drying chamber monitoring
- UV-C LED module for sterilization
- DC heating element and fan for controlled airflow drying
- Semi-automated packaging mechanism with QR code printer/sticker module
- Companion mobile application with Bluetooth/WiFi for remote monitoring

Methodology & Process for Implementation

- Design and fabrication of the solar-powered drying chamber
- Integration of hybrid power system (solar + battery)
- Sensor-based moisture monitoring and auto-cutoff logic
- UV-C sterilization cycle integration with safety interlock
- Companion app development for remote monitoring and notifications
- Packaging mechanism design with QR-based batch traceability
- Field testing and calibration with artisan user groups

System Architecture – Three Layers

LAYER 1 PHYSICAL DRYING UNIT	Solar-Hybrid Power Module – panel, battery, charge controller	Drying Chamber – heater, fan, UV-C sterilization
LAYER 2 SENSING & CONTROL	Microcontroller (ESP32) with Temperature/Humidity Sensors	Moisture-Triggered Auto-Cutoff Logic
LAYER 3 CONNECTIVITY & OUTPUT	Companion Mobile App – status, notifications, multilingual voice UI	Packaging & QR Traceability Module

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FEASIBILITY AND VIABILITY

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Analysis of Feasibility

- Core components (solar panel, battery, microcontroller, sensors, UV-C LEDs) are widely available, low-cost, and commonly used in student hardware builds
- Agarbatti manufacturing is a well-established home-based MSME sector with a large existing artisan base, giving a clear and reachable user group
- Rural solar adoption and smartphone penetration (for the companion app) have grown significantly, supporting real-world deployment

Potential Challenges & Risks

- UV-C exposure safety requires a fully enclosed chamber with interlock to prevent accidental exposure
- Battery cost and lifespan may affect long-term affordability for artisans
- Calibrating moisture sensors accurately across different humidity/climate conditions
- Ensuring the companion app remains simple enough for low digital literacy users

Strategies to Overcome

- Enclosed UV-C chamber design with automatic shutoff when opened, following standard UV-C safety practices
- Use of low-cost, readily available battery types with a modular, replaceable design
- Field calibration and testing across multiple regions/climates before final deployment
- Voice-first, icon-based companion app design requiring minimal text input

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IMPACT AND BENEFITS

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Potential Impact on the Target Audience

- Removes weather-dependency from agarbatti production, enabling consistent year-round output for rural women artisans
- Improves hygiene and product quality through UV-C sterilization, opening access to higher-value/export markets
- Reduces manual monitoring burden through the companion app, freeing up artisan time

Benefits of the Solution

- Social – reduces drudgery and health exposure (dust, contamination) for women artisans, improving working conditions
- Economic – consistent, higher-quality output increases earning potential and market access
- Environmental – solar-powered operation reduces reliance on conventional energy and lowers the carbon footprint

SolKiran vs Traditional Open-Air Drying vs Basic Electric Dryer

Parameter	SolKiran	Traditional Open-Air Drying	Basic Electric Dryer
Weather dependency	Hybrid solar+battery, independent	Fully weather-dependent	Grid-dependent, ongoing cost
Hygiene	UV-C sterilized	Exposed to dust/insects	No sterilization
Monitoring	Remote app notifications	Constant manual checking	Manual timer only
Operating cost	Solar-powered, low running cost	Free but inconsistent	Ongoing electricity cost

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RESEARCH AND REFERENCES

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